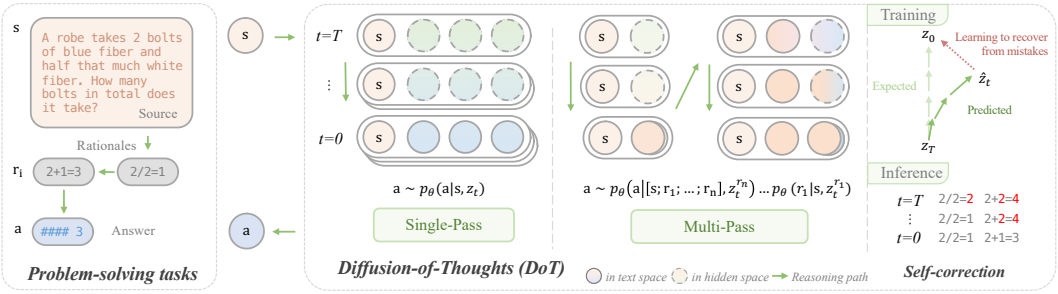

Figure 2: Demonstration of DoT pipeline. DoT diffuses all possible thoughts across diffusion timestep t. Multi-pass DoT disentangles each rationale and introduces causal bias. The stacked circles stand for the marginalization over other potential reasoning paths, which is implicitly carried out during the training of diffusion models.

We further propose a multi-pass (MP) variant of DoT, denoted as DoT^{MP}, which generates rationales in a thought-by-thought paradigm. This method disentangles the generation of multiple rationales and introduces casual inductive bias such that later rationale can be guided by stronger condition signals of prior rationales during the generation. Specifically, in the first pass, we generate the first rationale by $\mathbf{r}_1 \sim p_\theta^{DoT}(\mathbf{r}_1|\mathbf{s}, \mathbf{z}_t^{r_1})$, where $\mathbf{z}_t^{r_1}$ is the noised vector representation of $\mathbf{r}_1$ in diffusion model. Then $\mathbf{r}_1$ is connected to $\mathbf{s}$ as the condition $[\mathbf{s}; \mathbf{r}_1]$ to get $\mathbf{r}_2 \sim p_\theta^{DoT}(\mathbf{r}_2|[\mathbf{s}; \mathbf{r}_1], \mathbf{z}_t^{r_2})$, and then we have $[\mathbf{s}; \mathbf{r}_1; \mathbf{r}_2]$. Through multiple iterations, we can get the final answer: $\mathbf{a} \sim p_\theta^{DoT}(\mathbf{a}|[\mathbf{s}; \mathbf{r}_1; ...; \mathbf{r}_n], \mathbf{z}_t^{r_n})$.

3.3 Training

Scheduled sampling Diffusion models have intrinsic self-correcting capability through the multi-step denoising process. To further improve their self-correcting ability, we design a *scheduled sampling* [3] mechanism tailored for diffusion models such that self-generated error thoughts in previous timesteps are exposed and corrected during the training stage. Formally, for any timesteps s, t, u that satisfy $1 < s < t < u < T$, $\mathbf{z}_t$ is sampled from the forward distribution $q(\mathbf{z}_t \mid \mathbf{z}_0)$ in the training stage while during inference it is sampled from $q(\mathbf{z}_t \mid \mathbf{z}_\theta(\mathbf{z}_u; u))$ instead, where $\mathbf{z}_\theta$ is a denoiser neural network that reparameterizes $\mathbb{E}_q[\mathbf{z}_0|\mathbf{z}_t]$. The presence of such exposure bias may impede the model's ability to recover from erroneous thoughts during the generation process as the model $\mathbf{z}_\theta$ has only been trained on corruptions $\mathbf{z}_t$ diffused from oracle data. To mitigate this problem, we mimic the inference stage with probability ϵ_i during training depending on the current training step i, and ϵ_i linearly decays from 1 to ϵ_{min}. Specifically, for time-step t, we randomly sample a former time-step $u \in \{t+1, \ldots, T\}$, obtain $\mathbf{z}_u$ by forward noising and perform a model forward pass to get a predicted $\hat{\mathbf{z}}_0 = \mathbf{z}_\theta(\mathbf{z}_u; u))$. $\mathbf{z}_t$ is then sampled from $q(\mathbf{z}_t \mid \hat{\mathbf{z}}_0)$ to replace the regular one in loss calculation. Compared with scheduled sampling for autoregressive models, such a mechanism in DoT helps the model to recover from errors by considering global information instead of relying on the left-side tokens.

Coupled sampling In DoT^{MP}, correct previous thoughts are given in the training stage, which is not given during inference. Similar to auto-regressive decoding, DoT^{MP} may suffer from error accumulation during the thought-by-thought generation process. To enhance the self-correction ability of DoT^{MP}, we propose a coupled sampling mechanism by adding noise not only to the current thought but also to previous thoughts during training with some probability. For instance, the previous sequence $\mathbf{z}_0 = \textsc{Emb}([\mathbf{s}; \mathbf{r}_1])$ will be modified to $\mathbf{z}_0 = \textsc{Emb}([\mathbf{s}; \mathbf{r}_1; \mathbf{r}_2])$, with the partial noise being applied to $[\mathbf{r}_1; \mathbf{r}_2]$ rather than just the last rationale $\mathbf{r}_2$. Therefore, the model learns to be robust to errors in $\mathbf{r}_1$ when predicting $\mathbf{r}_2$, which better aligns with the inference stage. The new $\mathbf{z}_0$ will be reparameterized into $\mathbf{z}_t$ as before and other procedures keep the same.

Training objective Given a set of training data for problem-solving tasks of size D: $\{\mathbf{s}^j, \mathbf{r}^j_{1\ldots n}, \mathbf{a}^j\}_{j \in D}$, we have two training settings for DoT models: one is training from scratch, while the other is fine-tuning from the pre-trained diffusion model. In both training settings, we share the same training objective. For example, the objective is to minimize the negative

variational lower bound $\mathcal{L}_{\text{VLB}}(\mathbf{w}^z)$ in continuous diffusion models:

$$\mathcal{L}_{\text{VLB}}(\mathbf{w}^z) = \mathbb{E}_{q(\mathbf{z}_0|\mathbf{w}^z)}\left[\underbrace{\log \frac{q(\mathbf{z}_T|\mathbf{w}^z)}{p_\theta(\mathbf{z}_T)}}_{\text{Prior loss}} + \underbrace{\mathcal{L}_{\text{VLB}}(\mathbf{z}_0)}_{\text{Diffusion loss}} \underbrace{-\log p_\theta(\mathbf{w}^z|\mathbf{z}_0)}_{\text{Rounding loss}}\right], \tag{1}$$

where the rounding loss regularizes the embedding learning and the diffusion loss sums up the KL divergence of each time step t with different weighting terms. Please refer to Appendix A for a detailed training objective formulation of continuous and discrete diffusion models.

3.4 Inference

One of the significant advantages of diffusion models is their inference flexibility. Naturally, more complex problems may necessitate increased computation in reasoning time [2, 54], which can be controlled by setting a larger backward timestep T in DoT. However, continuous diffusion such as Plaid usually requires more timesteps, e.g., 4096 [18], to converge. To accelerate the sampling process of the continuous diffusion, we adapt the ODE Solver [38, 39] into a conditional form to fit the conditional training process (detailed in Appendix A.4). Moreover, sharing a similar idea of MBR [30], self-consistency [52] boosts the performance of CoT significantly by generating and aggregating multiple samples. In the context of diffusion models, we can also expect its potential improvement using self-consistency, thanks to their ability to naturally produce diverse responses [15]. After sampling m times to obtain multiple reasoning pathways $(\mathbf{r}_{i;1\ldots n}, \mathbf{a}_i)$ from DoT, self-consistency involves marginalizing over $\mathbf{r}_{i;1\ldots n}$ by taking a majority vote over $\mathbf{a}_i$, i.e., $\arg\max_{\mathbf{a}} \sum_{i=1}^{m} \mathbb{1}(\mathbf{a}_i = a)$. We consider this as the most "consistent" answer among the candidate set of m answers.

4 Experiments

We conduct experiments on both simple multi-digit multiplication and boolean logic reasoning as well as complex grade school math problems, to explore the reasoning paradigm in diffusion models.

4.1 Experimental Setup

Datasets and Metrics. Following Deng et al. [7], we employ the four-digit (4×4) and five-digit (5×5) multiplication problems from the BIG-bench benchmark [49], known to be challenging for LLMs to solve without CoT. Given that arithmetic reasoning is just one type of the reasoning ability, we also incorporate a boolean logical reasoning task [68]. For more complex tasks, grade school math problems require both language understanding and mathematical reasoning, so we adopt the widely-used GSM8K dataset [6]. We use the augmented training data from Deng et al. [7] and keep all original test sets unchanged. The statistics are listed in Appendix B.1. For both datasets, we use accuracy to measure the exact match accuracy of predicting the final answer, and throughput to measure the number of samples processed per second (it/sec) during inference with a batch size of 1.

Base Models. When training from scratch, we follow DiffuSeq[2] to use a 12-layer Transformer [51] encoder with similar size as GPT2-small (124M). We also use Plaid[3] (1.3B) [18], SEDD-small[4] (170M) and SEED-medium (424M) [37] as pre-trained diffusion language models for further fine-tuning. Both Plaid and SEDD are pre-trained on OpenWebText [10, 13], which is similar to that in GPT2, and the pre-training perplexity of Plaid and SEDD-small is on par with GPT2-small.

Baselines. We consider **Answer-only** and **CoT** as reasoning paradigms for comparison. Another important baseline is **Implicit CoT** [7], which distills thoughts into transformer layers to accelerate CoT reasoning. We use GPT-2 [4] at various scales (i.e., small 124M, medium 355M, and large 774M) as model baselines, known as conventional autoregressive language models. We mainly consider fine-tuning the model due to the relatively small model size, but we also consider prompting the strong commercial LLM **ChatGPT** `gpt-3.5-turbo-1106` using CoT few-shot demonstrations for completeness. We use 5-shot in the few-shot setting.

[2] `https://github.com/Shark-NLP/DiffuSeq`

[3] `https://github.com/igul222/plaid`

[4] `https://github.com/louaaron/Score-Entropy-Discrete-Diffusion`

Table 1: The main results on different problem-solving reasoning tasks. **Acc** (↑) is to measure the exact match accuracy of the predicted final answer. **Throughput** (↑) measures the number of samples processed per second during test with batch size equals to 1. The baseline results for Mult. and GSM8K datasets are taken from the implicit CoT paper [7] and have been validated for reproducibility by us. Bracketed numbers indicate the self-consistency results.

Models	4×4 Mult.		5×5 Mult.		Boolean logic		GSM8K-Aug	
	Acc	Throughput	Acc	Throughput	Acc	Throughput	Acc	Throughput
Answer-only								
GPT2-small	28.7	13.2	1.2	11.1	98.8	16.2	13.3	24.7
GPT2-medium	76.2	7.0	1.9	5.9	100	9.6	17.0	9.1
GPT2-large	33.6	4.8	0.9	4.0	100	7.4	12.7	9.1
ChatGPT (few-shot)	2.2	1.0	0.0	1.4	67.6	0.5	28.1	1.8
Chain-of-Thoughts (CoT)								
GPT2-small	100	2.3	100	1.5	100	0.8	39.0 (41.6)	2.0
GPT2-medium	100	1.2	100	0.8	100	0.5	43.9	1.1
GPT2-large	100	0.8	99.3	0.6	100	0.3	44.8	0.7
ChatGPT (few-shot)	42.8	0.1	4.5	0.1	75.8	0.2	61.5	0.2
Implicit CoT								
GPT2-small	96.6	8.9	9.5	7.9	-	-	20.0	16.4
GPT2-medium	96.1	4.8	96.4	4.3	-	-	21.9	8.7
Diffusion-of-Thoughts (DoT)								
From-scratch	100	62.5	100	61.8	100	55.2	4.6	22.7
Plaid	100	24.3	100	21.3	100	10.2	32.6 (36.3)	0.3
SEDD-small	100	59.2	100	55.5	100	33.3	45.3 (51.8)	1.0
SEDD-medium	100	31.8	100	28.5	100	17.2	53.5 (59.4)	0.5
Diffusion-of-Thoughts (DoT^{MP})								
From-scratch	100	11.8	100	9.5	100	3.7	5.5	8.6
Plaid	100	4.3	100	3.9	100	1.0	37.7	0.1
SEDD-small	100	9.9	100	9.2	100	3.3	43.2	0.2
SEDD-medium	100	4.5	100	4.0	100	1.7	53.3	0.1

Implementation Details. During tokenization, we treat all the digits as individual tokens. For DoT^{MP}, we append a special token <EOS> to the last thought, so when the model generates a thought followed by <EOS>, it stops generating further, which enables the model to decide the number of rationales dynamically. We conduct all the experiments on 8 NVIDIA V100-32G GPUs. During training, we set ϵ_{min} to be 0.95 as we find decreasing the probability of oracle demonstration hinders model training. We choose coupled sampling $\gamma = 0.01, k = 1$ and self-consistency $m = 20$. Following Plaid, we also adopt self-conditioning [5] during training. During inference, we set both the temperature of the score and output logit to 0.5 to sharpen the predicted output distribution while maintaining the ability to generate diverse samples. The sampling timesteps T is dynamic. By default, we set it to be 64. Considering that simple tasks do not necessitate an excessively large number of steps, we opt to reduce T while ensuring there is no notable performance drop. Other details are in Appendix B.3.

4.2 Results on Digit Multiplication and Boolean Logic

We first train DoT for digit multiplication tasks and a boolean logical reasoning task as the preliminary investigation, as shown in the left part of Table 1. We observe that neither ChatGPT nor the distilled Implicit CoT model can reach 100% accuracy. GPT-2 can be fine-tuned to achieve high accuracy but sacrifices throughput during CoT. Interestingly, DoT can attain 100% accuracy for these tasks while maintaining significant throughput with diffusion sampling steps set at 1 for multiplication datasets and 2 for the boolean logical dataset, achieving maximum $27\times$ speed-up compared to GPT-2. This preliminary finding indicates that DoT performs well in modeling exact math computation or boolean logic reasoning and benefits from its computational efficiency.